Assignment 2 – GitHub Collaboration & Automation

Oct 29, 2026 · @iman

Field	Value
Group size	2
Chosen theme	To-Do List
Full names	Kenza Dassaine · Iman El Isamy
Student IDs	[KENZA 168566 · IMAN 157195]
Course codes & sections	[CSC 3324 / SECTION01.02]
GitHub usernames	heretosee134 (Kenza) · elisamyiman8-hub (Iman)
GitHub repository URL	github.com/heretosee134/todo-list-project
GitHub Pages URL	heretosee134.github.io/todo-list-project

1. Project & Role Distribution

We built a small Python To-Do List library and followed the **2-member path**: two comparable feature areas, each owned by one member, with at least 3 unit tests each.

Member	Feature	Code file	Test file
Kenza (heretosee134)	Feature A: add and remove tasks	`todo/manage.py`	`tests/test_manage.py`
Iman (elisamyiman8-hub)	Feature B: complete, filter and stats	`todo/status.py`	`tests/test_status.py`

Kenza created the public repository, added Iman as a collaborator, and set up the project structure (README, `.gitignore`, `requirements.txt`, `pytest.ini`, CI workflow, `docs/`). Both features share one task format: `{"title": str, "done": bool}`.

2. Branches, Commits & Contributions

Each member developed only on feature branches; no feature work was pushed to `main`.

Kenza — main feature branch `feature/manage-tasks`:

- `Add add_task and remove_task logic`
- `Add unit tests for task management`
- `Fill in unit tests for task management`
- Other branches: `setup-structure` (project setup), `status-kenza` (status line + conflict resolution), `update-pages` (final Pages change + Pages URL in README)

Iman — main feature branch `feature/task-status`:

- `Add task completion, status filter and completion percentage`
- `Move status feature and tests into todo and tests folders` (this push produced the failed CI run)
- `Fix completion percentage formula, validate input, reject completing done tasks`
- Other branches: `status-iman` (status line for the conflict), `add-pages` (GitHub Pages project page)

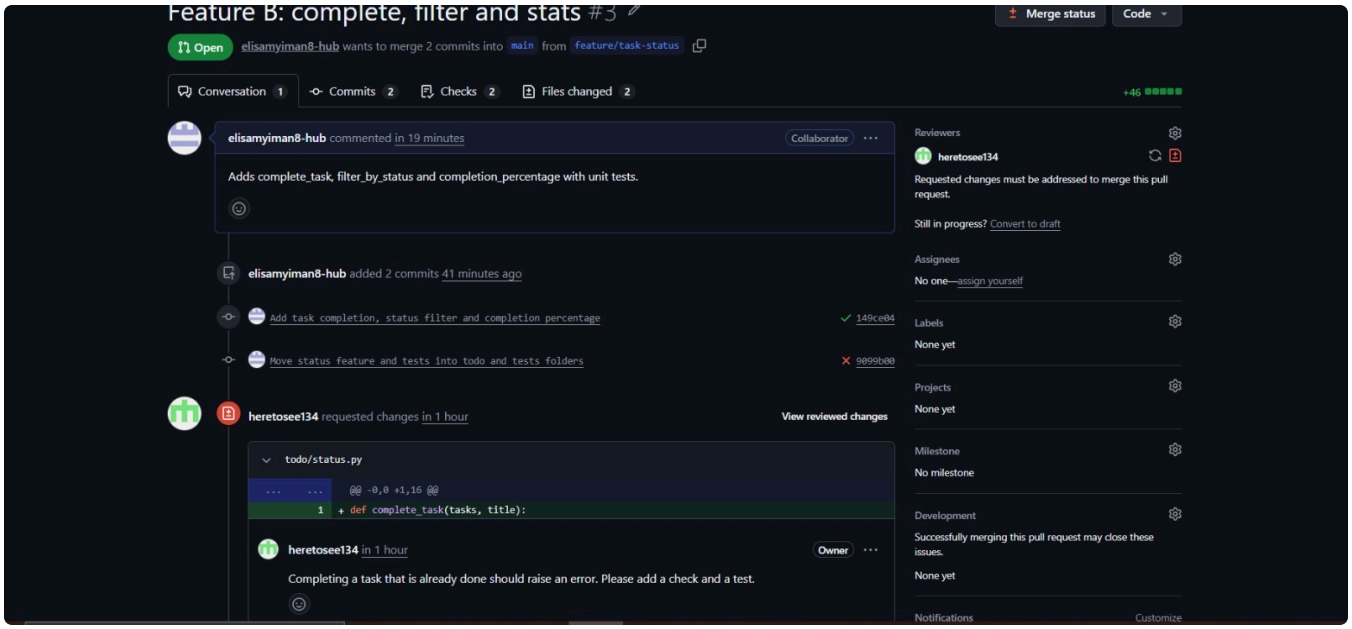
3. Pull Requests & Code Review

Every change reached `main` through a PR approved by the other member.

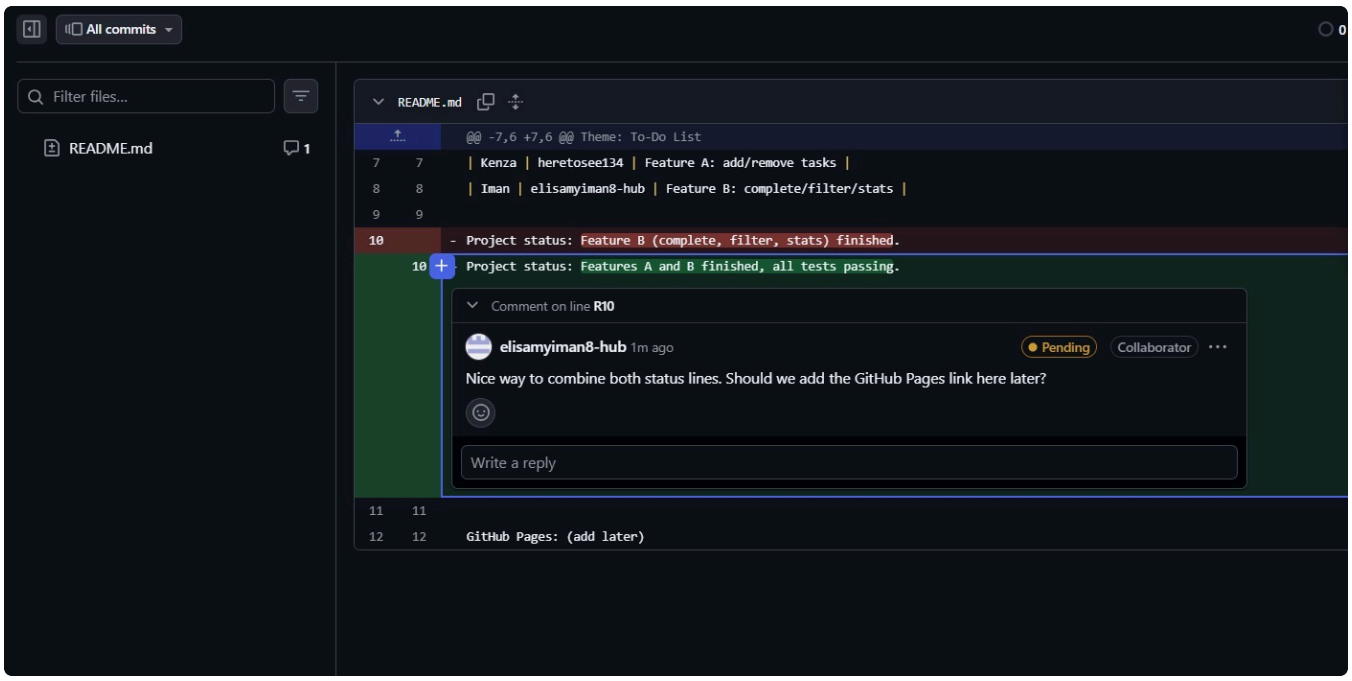
PR	Title	Author	Reviewer	Review outcome
#1	Feature/manage tasks	Kenza	Iman	Approved
#2	Set up project structure, README, CI workflow	Kenza	Iman	Comment, then approved
#3	Feature B: complete, filter and stats	Iman	Kenza	**Request changes**, then approved
#4	Update project status for Feature B	Iman	Kenza	Approved
#5	Update project status for Feature A (conflict PR)	Kenza	Iman	Line comment, then approved
#6	Add GitHub Pages project page	Iman	Kenza	Approved

PR	Title	Author	Reviewer	Review outcome
#7	Final project page update and Pages URL	Kenza	Iman	Approved

Request changes cycle (PR #3). In *Files changed*, Kenza left a line comment on `def complete_task` in `todo/status.py`: completing a task that is already done should raise an error. She submitted **Request changes**, which blocked the merge. Iman added the check (`ValueError: Task already completed`) and the test `test_complete_already_done_task_raises` on the **same branch** `feature/task-status`, committed, pushed, replied in the thread and re-requested review. The PR updated automatically; Kenza approved and it was merged.



Line comment by Iman (PR #5). On the resolved status line in `README.md`, Iman asked whether the GitHub Pages link should be added later; this was done in PR #7.



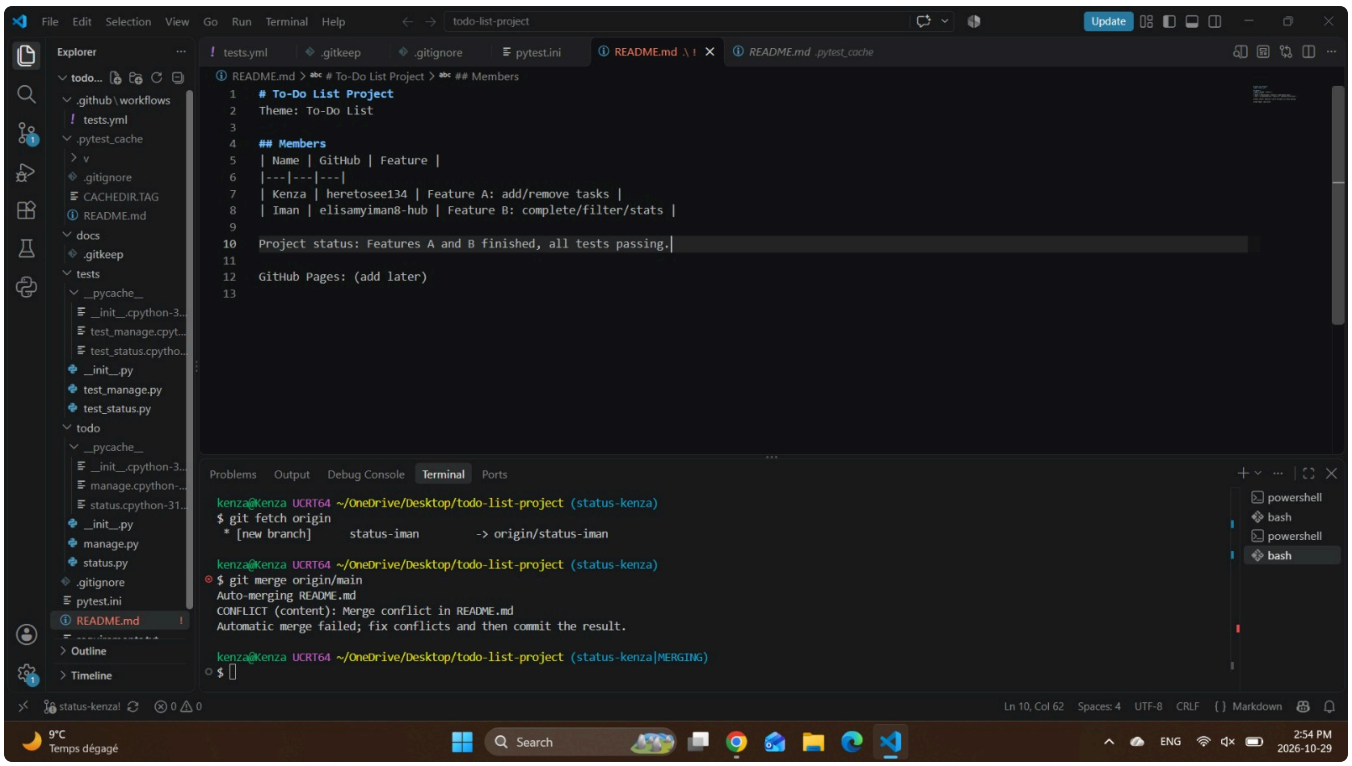
4. Merge Conflict

We created the conflict on the shared README line `Project status: Development in progress.`

1. Both members branched from the same `main`: Iman created `status-iman`, Kenza created `status-kenza`.
2. Iman changed the line to `Project status: Feature B (complete, filter, stats) finished.` (PR #4); Kenza changed it to `Project status: Feature A (add and remove tasks) finished.` (PR #5).
3. PR #4 was approved by Kenza and merged first. PR #5 then showed *This branch has conflicts that must be resolved* on `README.md`.
4. On `status-kenza`, Kenza ran `git fetch origin` and `git merge origin/main`, which stopped with `CONFLICT (content): Merge conflict in README.md`.

Current vs Incoming. *Current* (HEAD) was Kenza's branch with the Feature A line. *Incoming* was `origin/main` with Iman's merged Feature B line.

Resolution. Both lines were correct, so we combined them into `Project status: Features A and B finished, all tests passing.` Kenza removed the conflict markers, then ran `git add README.md`, `git commit -m "Resolve README project status conflict"` and `git push`. The conflict message disappeared, CI passed, and Iman reviewed and approved PR #5 before it was merged.



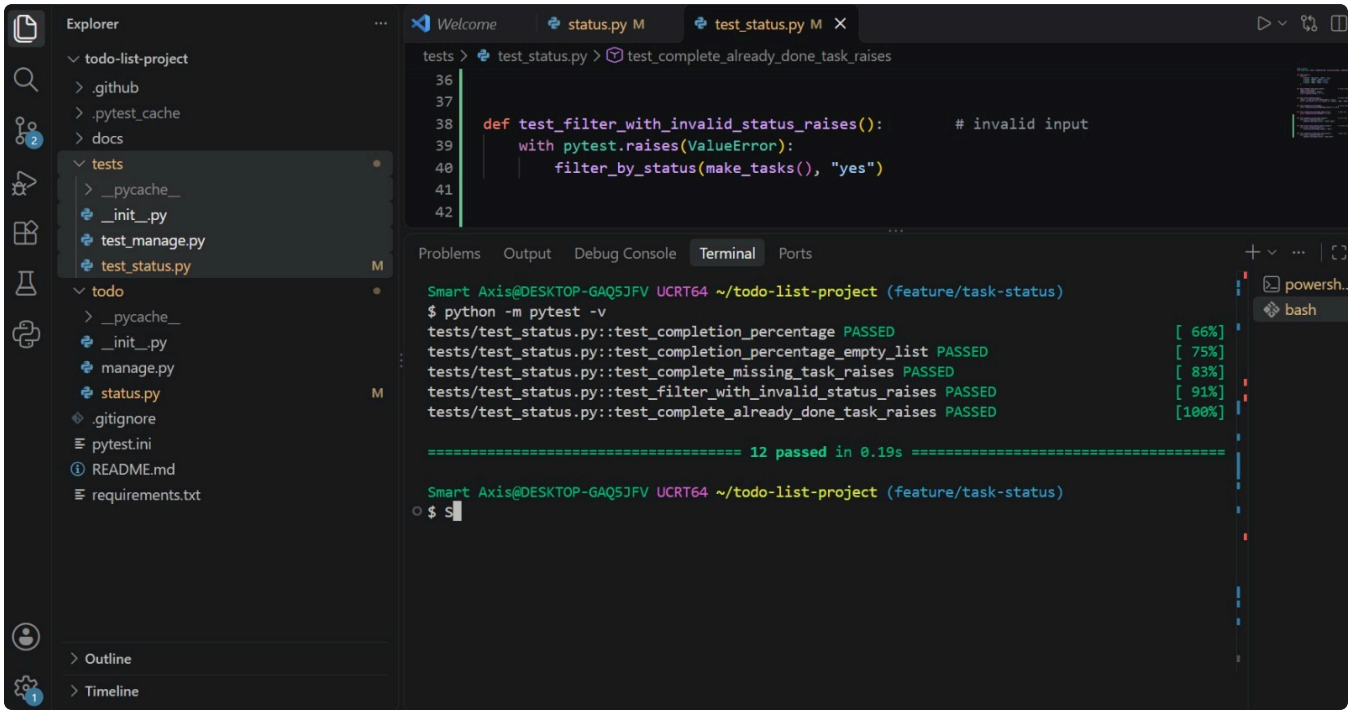
5. Unit Tests

The final suite has **12 pytest tests** (5 by Kenza, 7 by Iman), and all 12 pass locally (`python -m pytest -v` : *12 passed in 0.19s*) and in GitHub Actions.

Test	Owner	Type
`test_add_task_normal`	Kenza	Normal behavior
`test_remove_task_normal`	Kenza	Normal behavior
`test_add_task_strips_spaces_edge_case`	Kenza	Edge case
`test_add_task_empty_title_raises`	Kenza	Invalid input / error
`test_remove_missing_task_raises`	Kenza	Error case
`test_complete_task_marks_done`	Iman	Normal behavior
`test_filter_pending_tasks`	Iman	Normal behavior
`test_completion_percentage`	Iman	Normal behavior (1 of 4 done = 25.0%)
`test_completion_percentage_empty_list`	Iman	Edge case (empty list = 0.0)

Test	Owner	Type
`test_complete_already_done_task_raises`	Iman	Edge case (added after review)
`test_complete_missing_task_raises`	Iman	Error case
`test_filter_with_invalid_status_raises`	Iman	Invalid input (`done="yes"`)

Error cases use `pytest.raises(ValueError)`. Iman's tests build their own task lists, so they do not depend on Kenza's code.

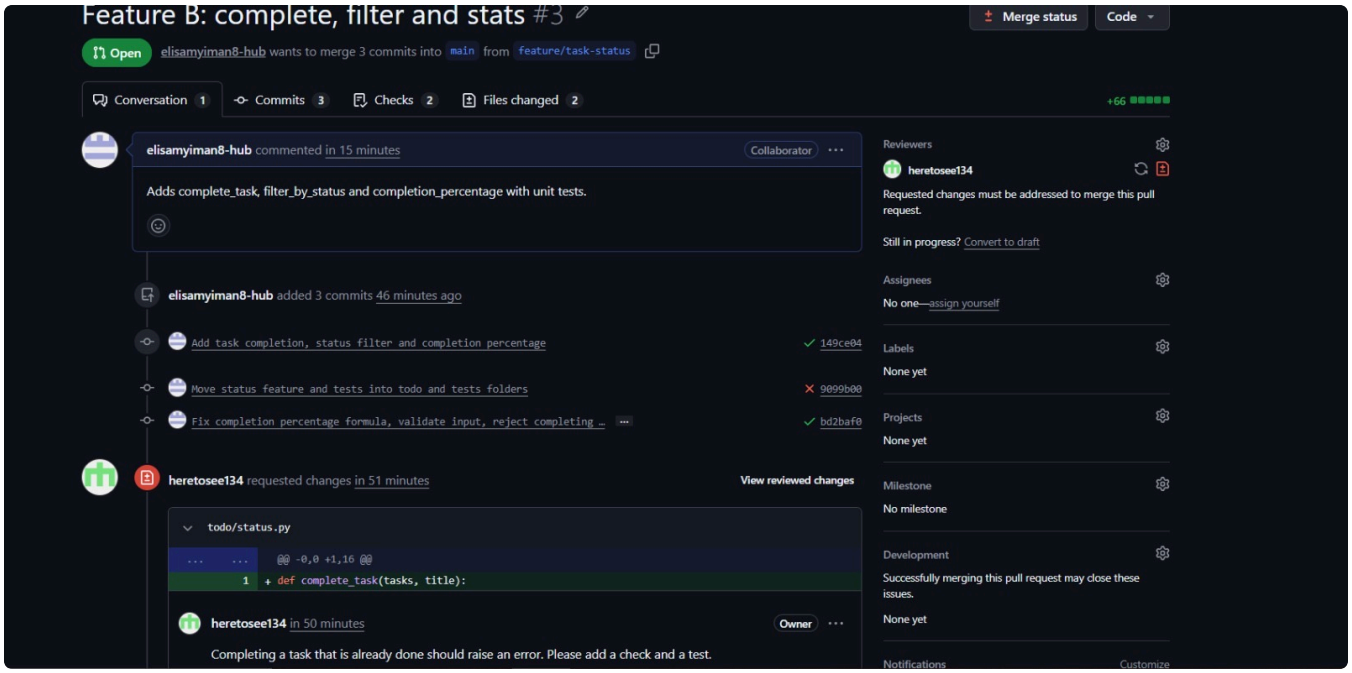


6. Continuous Integration

The workflow `.github/workflows/tests.yml` (name *Tests*) runs on every **push** and **pull_request**. Its single job `test` runs on `ubuntu-latest` with four steps: `actions/checkout` downloads the code, `actions/setup-python` installs Python 3.12, `pip install -r requirements.txt` installs pytest, and `python -m pytest -q` runs the whole suite.

Failed run. On PR #3, commit `9099b00` (*Move status feature and tests into todo and tests folders*) failed both checks (push and pull_request). The log showed `test_completion_percentage` failing with `assert 0.2 == 25.0`: `completion_percentage` returned a fraction because the `* 100` was missing.

Fix. Iman corrected the formula to `round(done_count / len(tasks) * 100, 1)`, added input validation, ran the suite locally (12 passed), then committed and pushed `bd2baf0`. The next run was green.

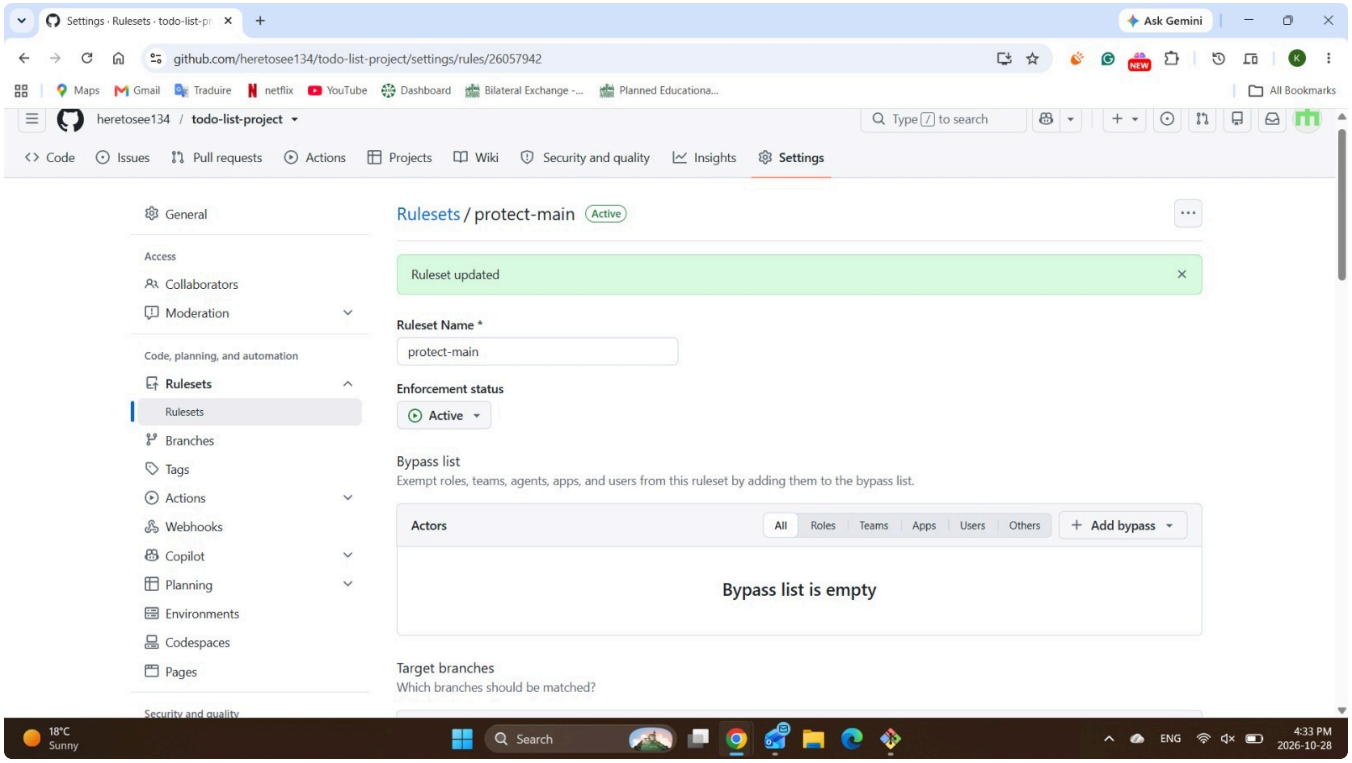


An earlier red run on PR #2 (`no tests ran`, exit code 5) happened only because the test folder was still empty.

7. Ruleset & Protected main

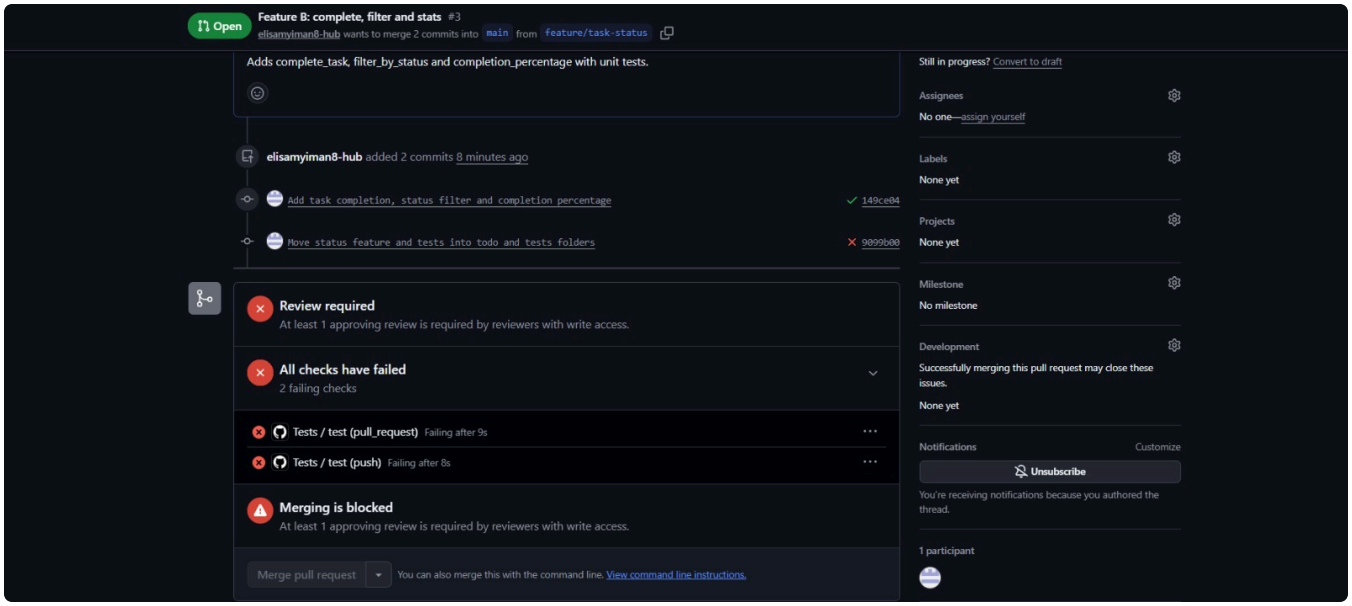
The active branch ruleset **protect-main** targets the default branch `main`, with an **empty bypass list**. Its rules:

- Restrict deletions
- Block force pushes
- Require a pull request before merging, with **1 required approval**
- Require status checks to pass: the `test` job from GitHub Actions (added once the workflow had run successfully)



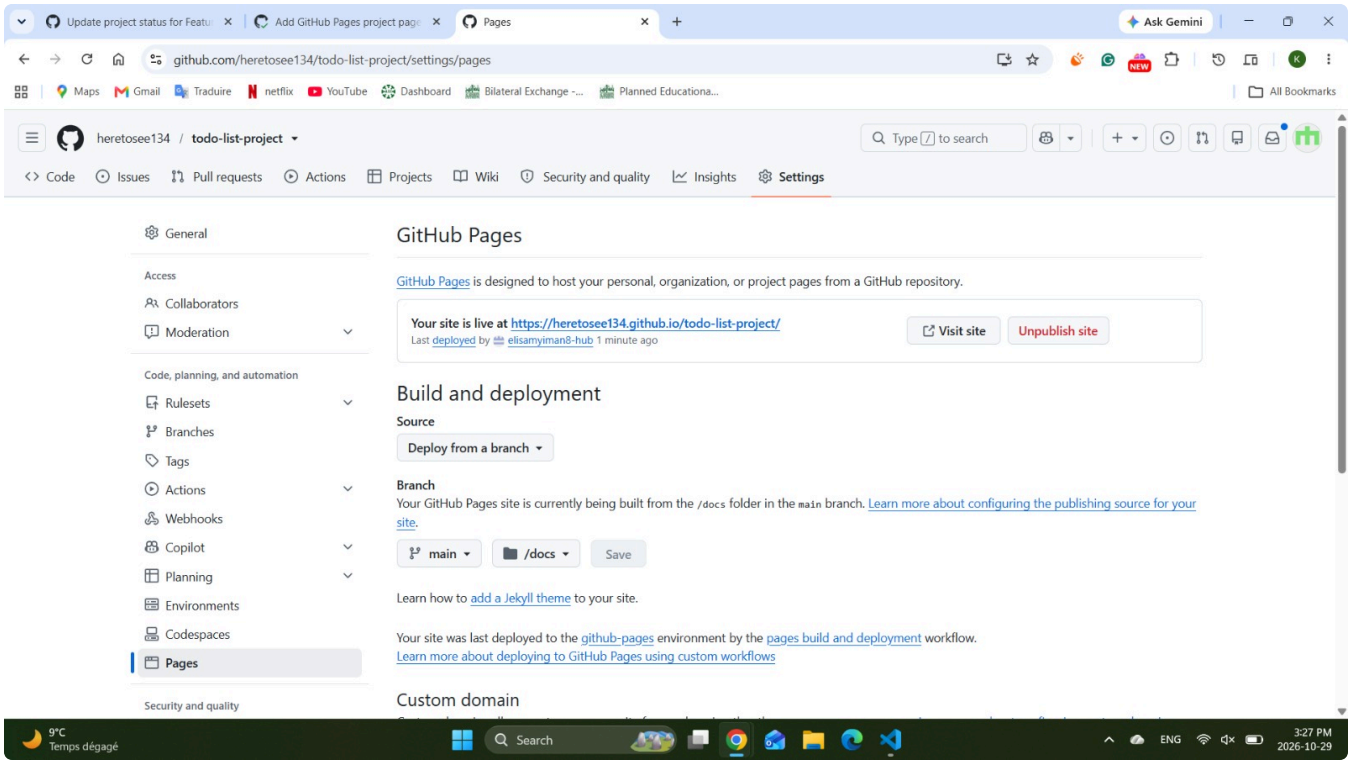
The protection worked from the start: Kenza's first `git push origin main` was rejected with *push declined due to repository rule violations — Changes must be made through a pull request*, so the setup went through PR #2 instead.

Before a PR could be merged, it needed **both** an approving review from the other member **and** a passing `test` check. PR #3 shows this: while CI was red and no approval existed, GitHub displayed *Review required*, *All checks have failed* and *Merging is blocked*, and the Merge button was disabled.



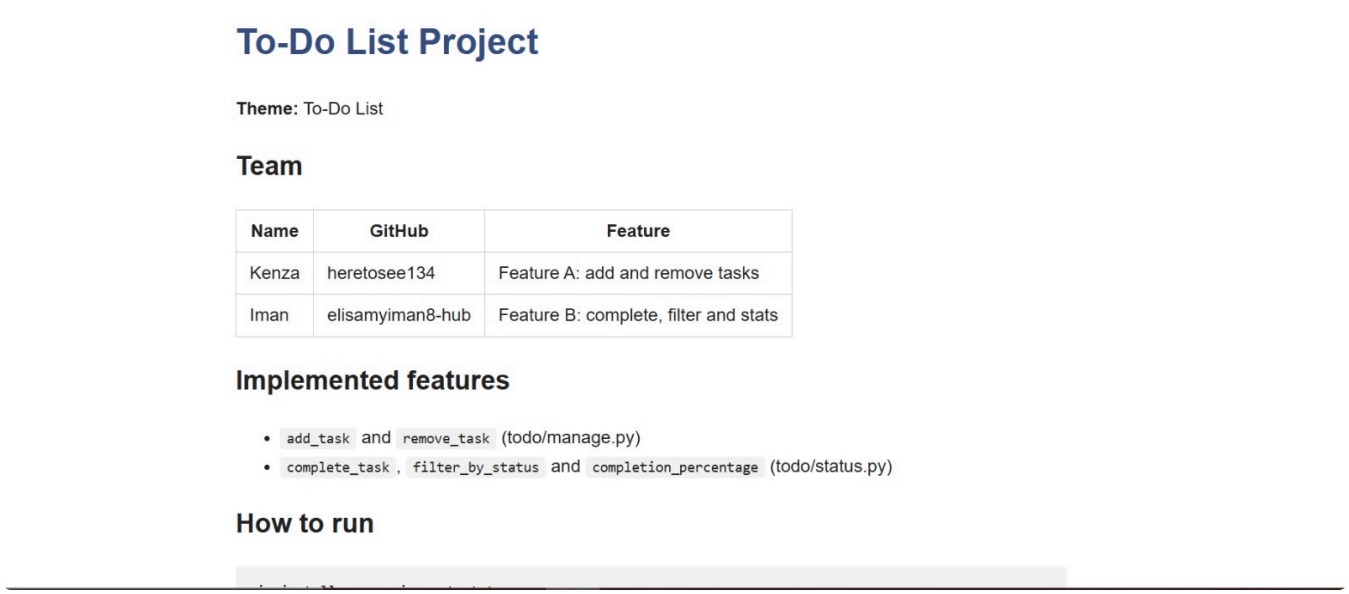
8. Continuous Deployment

The project page is live at heretosee134.github.io/todo-list-project. GitHub Pages is set to **Deploy from a branch**, branch `main`, folder `/docs`. After every merge into `main`, GitHub's *pages build and deployment* workflow republishes `docs/index.html` automatically.



The page (added by Iman in PR #6) shows the project title, theme, team members, implemented features and how to run the tests.

Final visible change. On branch `update-pages`, Kenza added the line *Version 1.0: all features tested with pytest and GitHub Actions.* under the theme, and replaced `GitHub Pages: (add later)` in the README with the live URL. Iman approved PR #7 and it was merged. About a minute later the new Version 1.0 line appeared on the live site without any manual deployment.



To-Do List Project

Theme: To-Do List

Team

Name	GitHub	Feature
Kenza	heretosee134	Feature A: add and remove tasks
Iman	elisamyiman8-hub	Feature B: complete, filter and stats

Implemented features

- `add_task` and `remove_task` (todo/manage.py)
- `complete_task`, `filter_by_status` and `completion_percentage` (todo/status.py)

How to run

To-Do List Project

Theme: To-Do List

Version 1.0: all features tested with pytest and GitHub Actions.

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Name	GitHub	Feature
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Implemented features

- `add_task` and `remove_task` (todo/manage.py)
- `complete_task`, `filter_by_status` and `completion_percentage` (todo/status.py)

How to run

9. Contribution Summary

Member	Feature	Branches	PRs authored	PRs reviewed	Tests
Kenza (heretosee134)	A: add and remove tasks	`setup-structure`, `feature/manage-tasks`, `status-kenza`, `update-pages`	#1, #2, #5, #7	#3 (Request changes, then approved), #4, #6	5 in `tests/test_manage.py`

Member	Feature	Branches	PRs authored	PRs reviewed	Tests
Iman (elisamyiman8-hub)	B: complete, filter and stats	`feature/task-status`, `status-iman`, `add-pages`	#3, #4, #6	#1, #2, #5 (line comment), #7	7 in `tests/test_status.py`

Kenza also owned the repository settings (collaborator, ruleset, GitHub Pages) and resolved the merge conflict; Iman produced and fixed the failed CI run.

10. Fair Use of AI

We used Claude (Anthropic's AI assistant) as a step-by-step guide. Our questions were mostly *"what should we do next, exactly?"* with screenshots of VS Code and GitHub, plus questions about errors such as the rejected push, a test file in the wrong folder, the "Deletion of directory failed" prompt and comment-only reviews. It also suggested starter code for `todo/status.py` and its tests, the workflow YAML, the `docs/index.html` page and a first draft of this report. We adapted it to our project (To-Do theme, our file layout, our real PR numbers), introduced the CI bug on purpose to demonstrate a failed run, and checked every step ourselves: tests run locally with pytest, CI results on GitHub, the PR and ruleset states, and the live Pages site. [Kenza: state here whether and how AI was used for Feature A.] Each member understands the code, tests, Git commands and reviews made under their account.

11. Sources & Reflection

Sources

- Course slides, *Session 2: Collaboration & Automation*
- GitHub Docs: repository access and collaboration
- GitHub Docs: rulesets
- GitHub Docs: GitHub Actions
- GitHub Docs: GitHub Pages
- pytest documentation
- Claude (AI assistant), as described in section 10

Reflection

Our hardest problem was keeping both local copies in sync with `main`: a branch created before the latest `git pull`, a file saved in the wrong folder and commands typed into the editor instead of the terminal all caused confusing results. The clearest example was a CI run that passed although the code had a bug, because `test_status.py` sat outside `tests/` and pytest never collected it. If a PR, merge, test or workflow behaves unexpectedly, we would first check the terminal prompt (right folder and branch), run `git status` and `python -m pytest -v` locally, and then read the failing step's log in the Actions tab.